BARATHKUMAR M

Electrical Engineer | Software Engineer | Embedded Systems Engineer | Automation Engineer
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Dindigul, Tamil Nadu, India

ABOUT ME

Final-year Electrical and Electronics Engineering student (Expected 2027) with expertise in Java, Python, Embedded C, PLC programming, and ESP32-based embedded development. Passionate about software development, embedded systems, IoT, and industrial automation, with hands-on experience through academic projects and internships. Strong analytical and problem-solving skills, committed to developing innovative, reliable solutions while continuously learning and contributing as a Software or Embedded Systems Engineer.

EDUCATION

NPR College of Engineering and Technology, Natham, Dindigul
B.E. Electrical & Electronics Engineering, Affiliated to Anna University, Current Academic Performance: 7.79 CGPA(UPTO 6th SEM) Expected2027

INTERNSHIP EXPERIENCE

PLC Automation Intern 05 Jul 2025-02 Aug 2025
Sri Vari Innovative Automation Solutions *T.Vadipatti Madurai District*

- Gained hands-on experience in plc panel wiring, electrical control-circuit troubleshooting, and industrial Automation.
- Complete done month of industrial training in PLC-based automation

Bhargave Rubber Pvt.Ltd.-Intern *05 Jun 2026-04 Jul 2026*
Bhargave Rubber Private Limited, *Madurai, Tamil Nadu*

- Gained practical exposure to industrial electrical systems, manufacturing processes, and plant operations.
- Observed and assisted with electrical troubleshooting, motor control, maintenance, and safety practices

PROJECTS

- **Smart Landmine Detection Robot |** Iot-enabled autonomous robotic system for remote landmine detection and hazardous -area monitoring using sensors and embedded control.
- **Sustainable Agriculture / Green Technology |** Developing smart, energy-efficient solutions integrating renewable energy iot embedded systems and automation for sustainable farming.
- **Smart Adaptive Road Intelligence & Rider Protection System |** Developed an intelligent embedded system to detect road Hazards in real time and provide immediate alerts to riders.
- **PLC-Based Motor Control & Automation System |** Developed an automated motor control system using plc Programming and ladder logic for reliable industrial motor operation
- **Inside Pipe Hydro Power Generation |** An innovative renewable energy project generating electricity from water Flowing inside pipelines, promoting sustainable and efficient power generation.

TECHNICAL SKILLS

Industrial Automation: PLC Programming Ladder Logic, HMI, Electrical Control Circuits

PLC Platforms: Omron & Siemens

Programming & Simulation: MATLAB, TINA Design, PLC Programming-Omron CX-Programmer

Embedded Systems : Arduino, ESP32, Respberry pi pico w

Electrical: power Systems, Relay Protection, Electrical Troubleshooting, Electrical Testing, Power Electronics, Renewable Energy Systems

Soft Skills : Teamwork, Problem Solving, Time Management, Leadership, Adaptability, Critical Thinking, Responsibility & Accountability.

CERTIFICATIONS

- MATLAB for Electrical Engineering Applications – AB Technologies
- Industrial Automation Using PLC – Shree Technologies, Coimbatore

ACHIEVEMENTS

- Won Third Prize in Paper Presentation at **SYSED AMMAL ENGINEERING COLLEGE** Ramanathapuram on "**Smart Landmine Detection Rover"** is an Iot based autonomous robotic system developed for defense and security application
- **Participated in Paper Presentation at-Government College of Technology Coimbatore" Smart Adaptive Road Intelligence & Rider protection System** is designed to detect road hazards in real time and alert riders and improving the safety
- **Participated in Science, Technology & Startup Fest- Smart Landmine Detection Rover** is an Iot based autonomous robotic system developed for defense and security application
- **Skill Development Program- "Industrial Automation Using plc"** this skill development program focuses on plc-based industrial automation for controlling and monitoring machines, improving productivity and process industries.
- **Skill Development Program- "MATLAB for Electrical Engineering Applications"** is used to model and analyze electrical machines such as motors and transformers
- **Participated in Paper Presentation at - NPR College of Engineering and Technology "Smart Adaptive Road Intelligence & Rider protection System"** is designed to detect road hazards in real time and alert riders and improving the safety.
- **Submitted Idea in MSME Idea Hackathon 5.0** on "Inside Pipe Hydro Power Generation," presenting an innovative approach to renewable energy and sustainable power generation solutions.

WORKSHOPS

- **Smartphone Debugging and Troubleshooting – NPR College of Engineering and Technology**, gained hands-on experience in identifying and resolving hardware and software issues in mobile devices